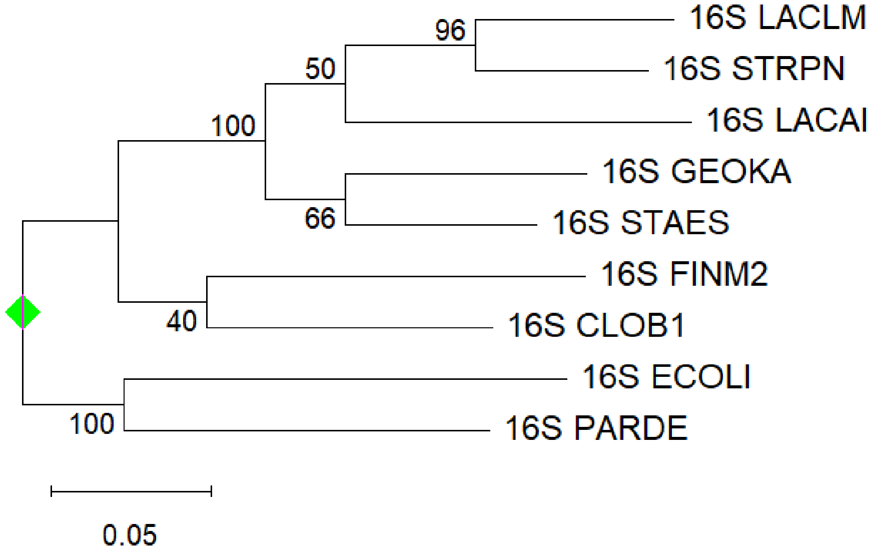
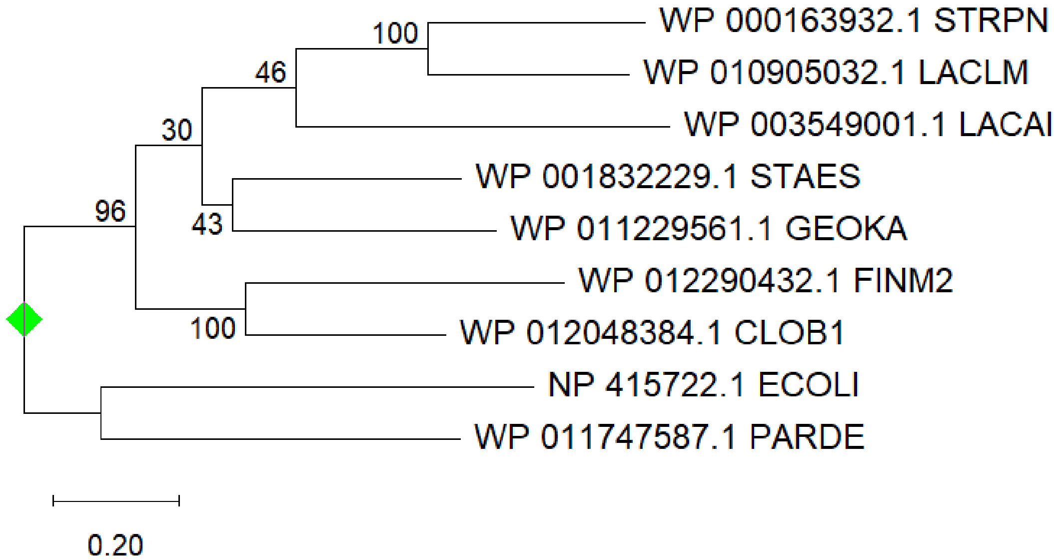
Trees

Task 1

((((16S_LACLM:0.0627231489254693,16S_STRPN:0.0546692802568959)0.9667[stddev=3
.57770876]:0.0408085180896199,16S_LACAI:0.108946903254917)0.5000[stddev=9.12870
929]:0.025046690314821,(16S_GEOKA:0.0759810590808846,16S_STAES:0.06067833447
20458)0.6667[stddev=8.64869932]:0.0250079139677451)1.0000[stddev=0.00000000]:0.04
63140337355838,(16S_FINM2:0.118939452876122,16S_CLOB1:0.0898518232815459)0.4
000[stddev=8.94427191]:0.0279772381502057,(16S_ECOLI:0.139403373074632,16S_PAR
DE:0.114977352436673)1.0000[stddev=0.00]:0.0618470895549977);



16S RNAs

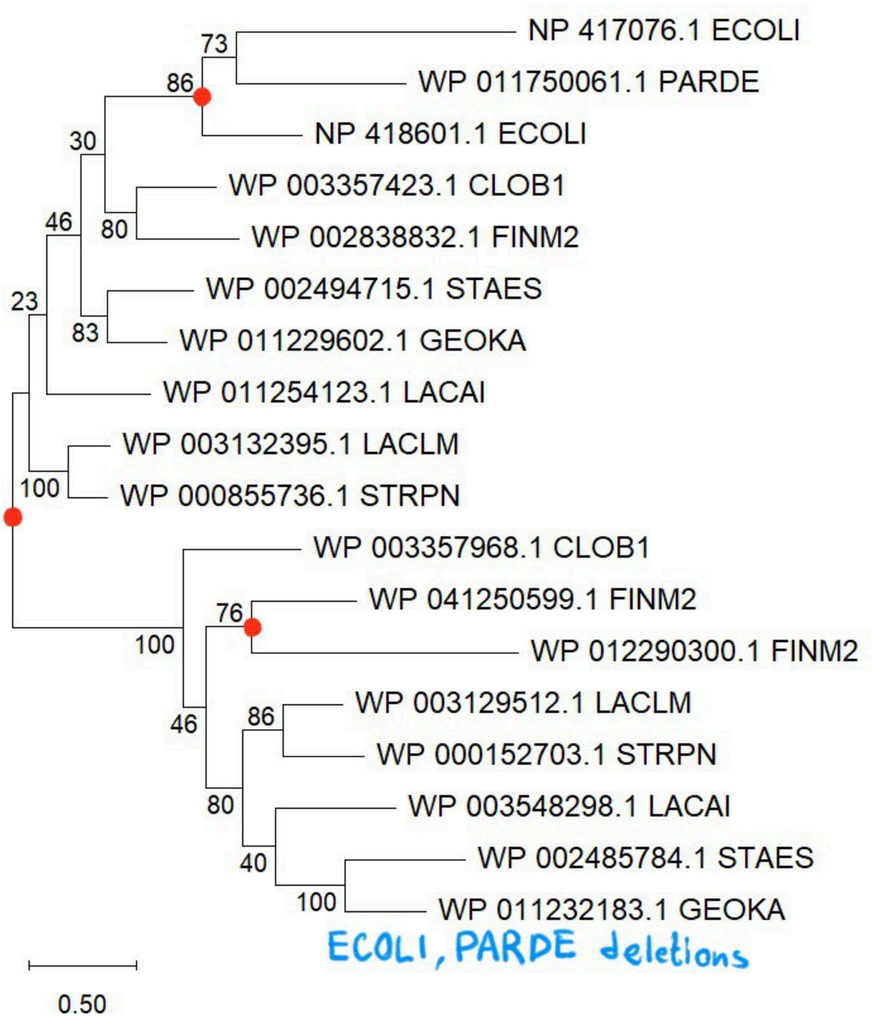


peptidyl-tRNA-hydrolases

The topologies look completely identical but they are a little bit different from gold standart phylogenetic species tree for this bacteria. The difference is in STAES/GEOKA separation node.

Task 2

FINM2 organism has the greatest number of methyltransferase genes (3).



Duplication events are specified with red dots.

The tree is separated into two large clades from one ancient duplication event. It follows from the fact that both of these clades contain almost all of our species, which means that duplication happened before their differentiation.

The only species which don't have paralogs in the second ancient clade from that duplication are ECOLI and PARDE, which indicates that they had deletion. It is further supported by the fact that ECOLI and PARDE are closest species to each other, so, probably, the deletion happened only once in their common ancestor, and haven't happened to any other species.

ECOLI/PARDE subclade in the upper clade had another one newer duplication. Similar for FINM2 in the lower clade.

All the other nodes are likely speciation events.

That means that each leaf from the upper clade is paralogous to each leaf in the lower clade as they diverge from duplication event. Inside those two large clades all the leaves can be called orthologous to each other within the clade, except for ECOLI and FINM2 cases which had additional duplication, so they are paralogs to each other.

Example of old duplication: the deep split between the upper and lower methyltransferase clades. Each contains copies from many species.
Example of new duplication: the FINM2-specific pair WP_041250599.1 and WP_012290300.1.